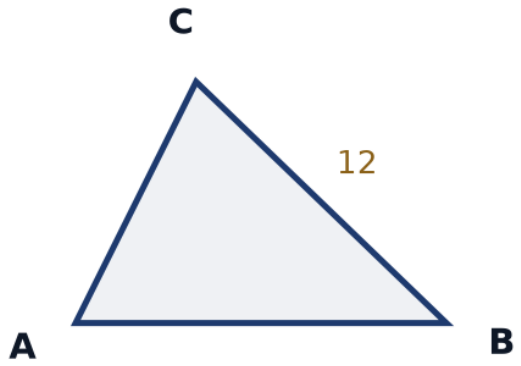
Law of Sines and Law of Cosines

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.04

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Law of Sines

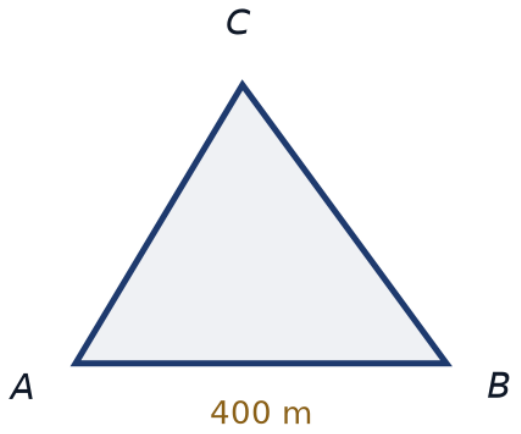
In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:



a angle C **b** side b (2 d.p.) **c** side c (2 d.p.)

2 [Foundational] • Target: Law of Sines

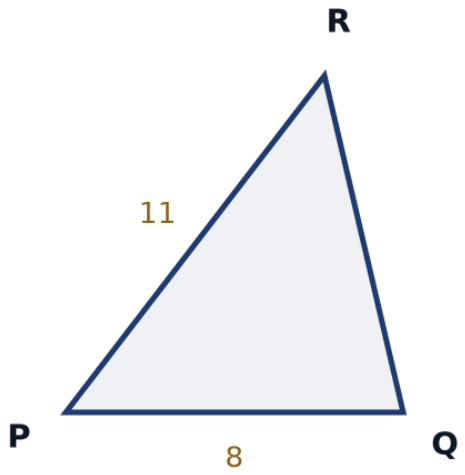
Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.



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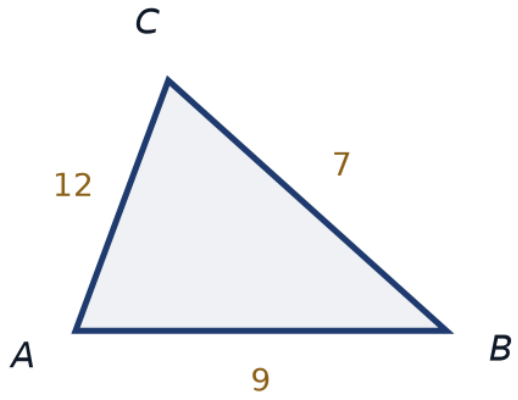
3 [Foundational] • Target: Law of Cosines

In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.



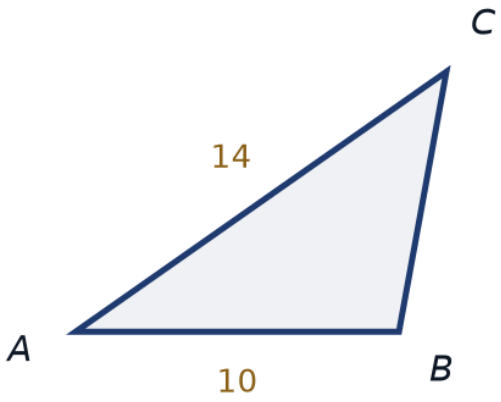
4 [Foundational] • Target: Law of Cosines

A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.



5 [Foundational] • Target: Law of Cosines

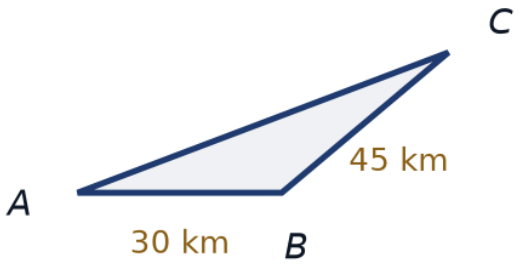
Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.



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6 [Foundational] • Target: Law of Cosines

A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.



7 [Foundational] • Target: The ambiguous case (SSA)

In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The ambiguous case (SSA)

In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

9 [Intermediate] • Target: Choosing the right law

For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

a Two angles and one side are known (AAS).

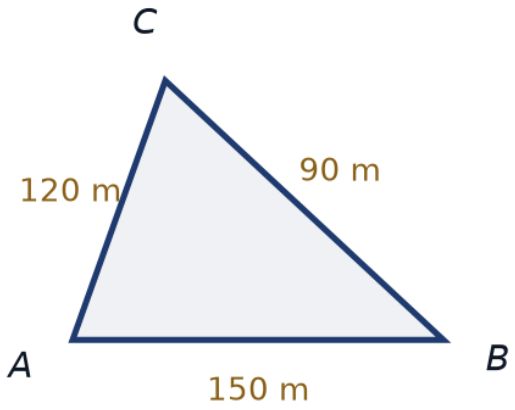
b Three sides are known (SSS).

c Two sides and the included angle are known (SAS).

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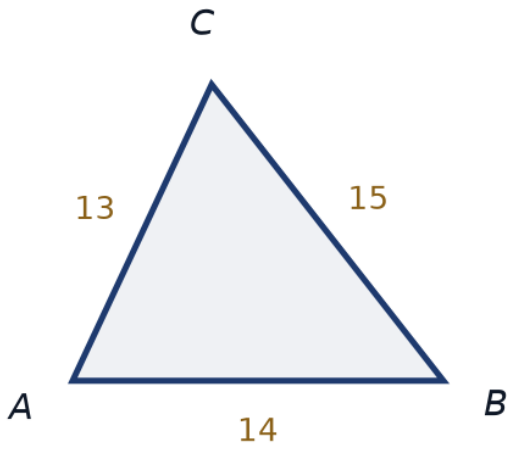
10 [Intermediate] • Target: Choosing the right law

A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.



11 [Intermediate] • Target: Heron's formula

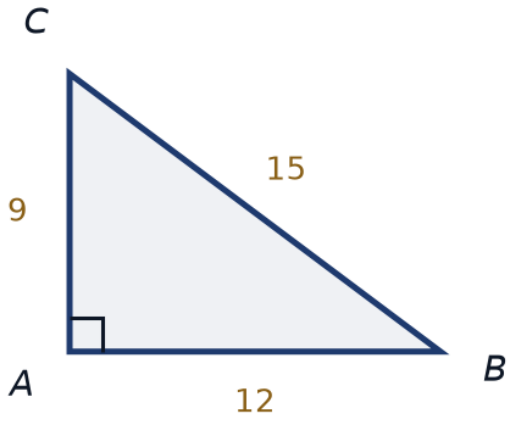
State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.



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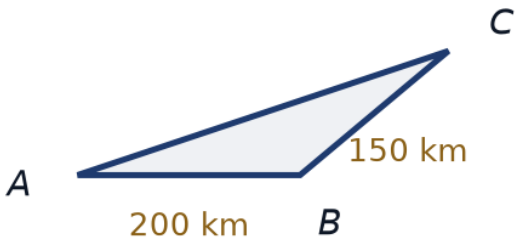
12 [Intermediate] • Target: Heron's formula

A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.



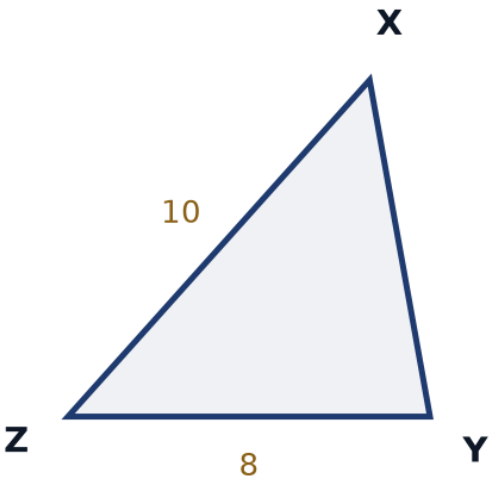
13 [Intermediate] • Target: Navigation and bearing problems

A plane flies on a bearing of $N\,60°\,E$ for 200 km, then changes course to a bearing of $S\,80°\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.



14 [Intermediate] • Target: Solving oblique triangles from a diagram

In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.



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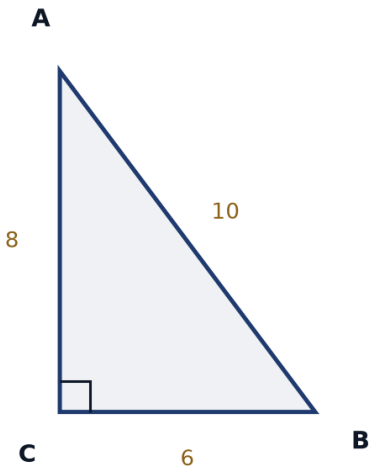
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Verifying triangle solutions with the angle sum** • **Cross-Lesson: 7.03 Exponential and Logarithmic Equations**

For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

16 **[Challenging]** • **Target: Choosing between multiple valid approaches** • **Cross-Lesson: 7.04 Applications: Growth and Decay**

A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.



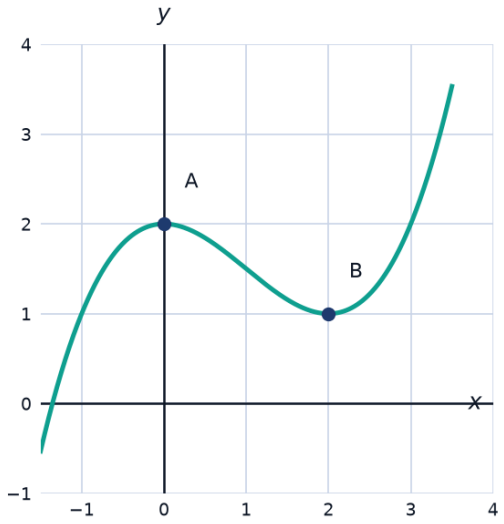
17 **[Challenging]** • **Target: Verifying triangle solutions with the angle sum** • **Cross-Lesson: 8.01 Rational Functions and Asymptotes**

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

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18 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 8.02 Graphing Rational Functions

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



19 [Challenging] • Target: Law of Sines • Cross-Lesson: 8.03 Introduction to Limits

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

20 [Challenging] • Target: Law of Cosines • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Two sides of a right triangle are 7 cm and 10 cm. If the third side is the hypotenuse, find: (a) the exact hypotenuse length, and (b) the measure of the smaller acute angle to the nearest tenth of a degree.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Law of Sines

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. 41.4°

b. 46.4°

c. 138.6°

d. 36.4°

22 [Foundational] • Target: Law of Cosines

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. 44.4°

d. 39.4°

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23 [Foundational] • Target: The ambiguous case (SSA)

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120^\circ$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. 14.8

c. 15.8

d. 12.2

24 [Foundational] • Target: Choosing the right law

In ΔABC, $a = 5$, $b = 7$, and $\angle C = 60^\circ$. Find side c (Law of Cosines).

a. $\sqrt{39}$

b. $1 + \sqrt{39}$

c. $\sqrt{74}$

d. 12

25 [Foundational] • Target: Heron's formula

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

a. 7.2

b. 10.8

c. 8.4

d. 9.6

26 [Foundational] • Target: Navigation and bearing problems

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

a. 6 units

b. 13 units

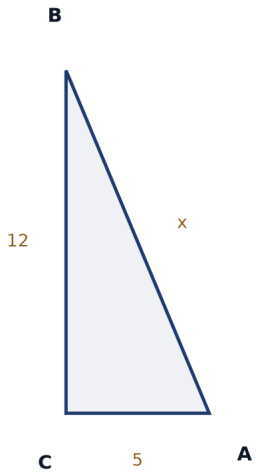
c. 36 units

d. 7.5 units

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27 [Foundational] • Target: Solving oblique triangles from a diagram

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. Given that AC = 5, BC = 12, and angle BAC = θ, find the value of $\sin(\theta)$.



a. $\frac{12}{13}$

b. $\frac{5}{13}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

28 [Foundational] • Target: Verifying triangle solutions with the angle sum

Use the sum formula to find $\sin(30^\circ + 60^\circ)$.

a. $\frac{11}{10}$

b. $\frac{1}{2} + \frac{\sqrt{3}}{2}$

c. -1

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Choosing between multiple valid approaches

In the diagram, ΔABC is a right-angled triangle with the right angle at C. The length of the hypotenuse AB is 26 units. Point D lies on AB such that CD is the altitude to the hypotenuse. If $AD = 4$ units and $BD = x$ units, find the value of x.

a. 12

b. 13

c. 16

d. 22

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30 [Intermediate] • Target: Law of Sines

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

a. 6 cm

b. 5 cm

c. 7 cm

d. 8 cm

31 [Intermediate] • Target: Law of Cosines

The diagram shows a right-angled triangle ABC, with the right angle at C. The altitude from C to the hypotenuse AB is CD. Given that $AD = 4$ and $DB = 9$, find the length of the altitude CD.

a. 3

b. 4

c. 5

d. 6

32 [Intermediate] • Target: The ambiguous case (SSA)

In a triangle, sides $a = 15$ and $b = 14$ meet at an included angle of 123°. Using the Law of Cosines, which expression gives side c?

a. $\sqrt{15^2 + 14^2 + 2(15)(14)\cos 123^\circ}$

b. $\sqrt{15^2 - 14^2 - 2(15)(14)\cos 123^\circ}$

c. $\sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$

d. $15^2 + 14^2 - 2(15)(14)\cos 123^\circ$

33 [Intermediate] • Target: Choosing the right law

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. 100

b. 110

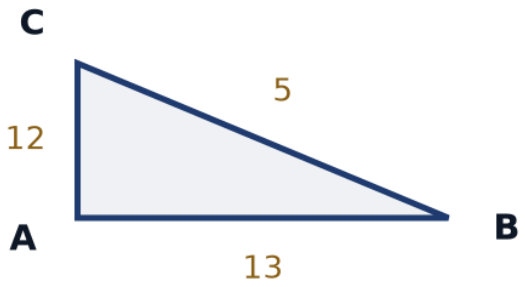
c. 10

d. 210

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34 [Intermediate] • Target: Heron's formula

In the right-angled triangle ABC shown below, the length of the side adjacent to angle A is 12 units and the length of the hypotenuse is 13 units. What is the cosine of angle A?



a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{5}{12}$

d. $\frac{13}{5}$

35 [Intermediate] • Target: Navigation and bearing problems

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

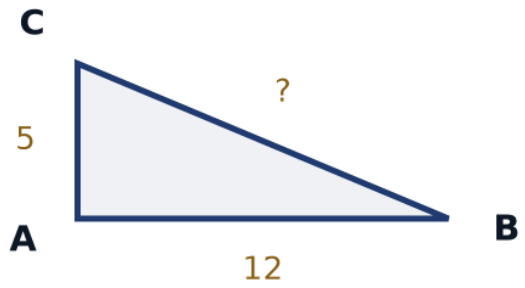
b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

36 [Intermediate] • Target: Solving oblique triangles from a diagram

A right triangle has legs 5 and 12. Hypotenuse is:



a. 11

b. 13

c. 17

d. 15

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37 [Intermediate] • Target: Verifying triangle solutions with the angle sum

Find the sum of the first 5 terms of the arithmetic series with $a_1 = 1$ and $d = 4$.

a. 45

b. 49

c. 40

d. 50

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 8.02 Graphing Rational Functions

Convert the polar point $(4, 30^\circ)$ to Cartesian coordinates.

a. $(2\sqrt{3}, 3)$

b. $(-2\sqrt{3}, -2)$

c. $(1 + 2\sqrt{3}, 2)$

d. $(2\sqrt{3}, 2)$

39 [Challenging] • Target: Law of Sines • Cross-Lesson: 8.03 Introduction to Limits

Given $f(x) = (x - 2)^2(x + 1)$, find the multiplicity of the root $x = 2$.

a. 3

b. 1

c. 2

d. 4

40 [Challenging] • Target: Law of Cosines • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{1}{2}$.

a. 17

b. 20

c. 18

d. 19

41 [Challenging] • Target: The ambiguous case (SSA) • Cross-Lesson: 9.02 Geometric Sequences and Series

Find the sum of the first 3 terms of the geometric series with $a_1 = -1$ and $r = 3$.

a. -13

b. -3

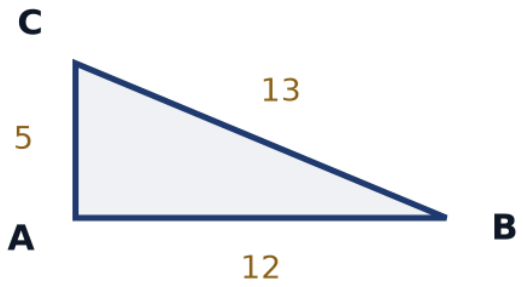
c. -12

d. -14

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42 [Challenging] • Target: Choosing the right law • Cross-Lesson: 9.03 The Binomial Theorem

A right triangle has legs 5 and 12 with hypotenuse 13. Find $\tan\theta$, where θ is the angle opposite the side of length 5.



a. $\frac{12}{13}$

b. $\frac{13}{5}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

43 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Functions and Function Notation

Use the difference formula to find $\cos(90^\circ - 30^\circ)$.

a. $-\frac{1}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{3}{5}$

d. $\frac{1}{2}$

44 [Challenging] • Target: Navigation and bearing problems • Cross-Lesson: 1.02 Domain and Range

Find the sum of the first 10 terms of the arithmetic sequence with $a_1 = 3$ and $d = 2$.

a. 130

b. 118

c. 21

d. 120

45 [Challenging] • Target: Solving oblique triangles from a diagram • Cross-Lesson: 1.03 Average Rate of Change

Find the sum of the first 5 terms of the geometric sequence with $a_1 = 2$ and $r = 3$.

a. 242

b. 162

c. 240

d. 245

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